CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

College of Computer Studies

Bachelor of Science in Information Technology

IT411 CAPSTONE 2 DELIVERABLE — STANDALONE EXTRACTION

PlayIT — Requirements Traceability Matrix (RTM v3.0)

Bidirectional Traceability Matrix (Findings F-01 to F-16), Test Coverage Table (§4.1), and Verification Protocol (§5)

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4. Requirements Traceability Matrix (RTM v3.0)

The following Requirements Traceability Matrix (RTM v3.0) demonstrates the bidirectional relationship linking **Empirical MVP Validation Findings & Stakeholder Evidence**, **Capstone Research Objectives**, **Refactored SRS Requirements**, **SDD v2.0 Architectural Components**, **Software Test Documents (STD) Test Cases**, and **Validation Verification Targets**.

Finding ID	Empirical Validation Evidence & Stakeholder Voice	Capstone Objective / SMART Goal	Refactored Req. ID	SDD v2.0 Design Component	Target STD Test Case(s)	Round 2 Field Verification Target	Priority
F-01	**PED-08 (50% flagged):** Pronunciation of letter sounds; Teacher T-1: *"M should be /m/ (mmm) rather than 'ma'."*	**Gen Obj 1 / HI-1** (Pure Phoneme Model)	**FR-02**	`AudioResolver`, `AudioPlayer` (`SoundPool`), `assets/audio/phonemes/ph_*.wav`; `AudioPlaybackManager` planned	**TC-AUD-01:** `SoundPool` start latency ≤ 50 ms **TC-AUD-02:** Zero schwa audio audit	**Instrument A (Part 1):** ≥ 3/4 teachers rate pure across 26 letters	**P0**
F-02	**Facilitator Notes:** Children hesitated at mic; perceived ease was lowest dimension (7 of 16 children did not choose the top face = 43.8%).	**Gen Obj 2 / SI-3** (Child Independence)	**FR-03**	`MicStatus`, `MicButton` (shipped); `MicStateVisualizer` with the `AudioRecord` RMS loop (planned)	**TC-MIC-01:** Tap-to-Listen ≤ 100 ms **TC-MIC-02:** Ripple scales with RMS input	**Instrument B / D-10:** Child hesitation drops to < 15%	**P1**
F-03	**Validation Gap:** MVP testing lacked empirical ASR agreement scoring against teacher judgments.	**Gen Obj 2 / SI-1** (ASR Agreement)	**NFR-ASR-01**	`SpeechValidator` (role of the planned `SayItJudge`), per-letter Vosk grammars	**TC-ASR-01 to 05:** Recognition across child pitch/noise ≤ 40 dB **TC-ASR-06:** False reject ≤ 15%	**Instruments B & C:** ≥ 80% agreement with teacher ratings	**P1**
F-04	**Validation Gap:** Speech latency unmeasured in field due to missing diagnostic logging.	**Gen Obj 2 / SI-2** (Immediate Feedback)	**NFR-PERF-01**	`TelemetryLogger`, Monotonic Clock (`elapsedRealtime`)	**TC-PERF-01:** Say It P90 ≤ 0.5 s **TC-PERF-02:** Find It P90 ≤ 0.3 s	**Instrument D (D-01, D-02):** Timed latency verification	**P1**
F-05	**Design Audit:** 5 CVC words (AIM, BEE, TOY, BOY, ZOO) contained unintroduced vowel teams/diphthongs.	**Gen Obj 4 / O4** (Decodable CVC Synthesis)	**FR-13**	`BLEND_IT_WORD_SEEDS` (`DatabaseModule`), rewritten on every app start	**TC-BLD-01:** 100% words match unlocked chapter letters	**Instrument A (Part 2):** Teacher decodability audit	**P1**

Finding ID	Empirical Validation Evidence & Stakeholder Voice	Capstone Objective / SMART Goal	Refactored Req. ID	SDD v2.0 Design Component	Target STD Test Case(s)	Round 2 Field Verification Target	Priority
F-06	**Teacher Feedback:** Teachers requested batch data export for classroom tracking; missing telemetry.	**Gen Obj 5 / O5** (Progress Tracking)	**FR-NEW-TEL**	`TelemetryLogger`, `CsvExportManager`	**TC-TEL-01:** Telemetry schema validation **TC-EXP-01:** PIN gate enforcement **TC-EXP-02:** Zero network calls	**Instrument C:** Successful CSV export on test tablet	**P1**
F-07	**PED-12 (4 of 4):** Teachers judged the app ready as a supplementary classroom tool; classroom stations share devices.	**Gen Obj 5 / O5** (Classroom Stations)	**FR-14**	`ProfileEntity`, `ProfileRepository`, Avatar Switcher	**TC-DB-03:** Multi-profile data isolation test	**Instrument D:** 6 profiles operate concurrently without data bleed	**P1**
F-08	**Field Observation:** Need formal proof that app survives unexpected OS kill without progress loss.	**Gen Obj 5 / O6** (Data Reliability)	**NFR-REL-01**	Room Transaction DAOs, Schema v4 Migration (planned)	**TC-DB-01:** 20 force-close crash tests **TC-DB-02:** Migration preservation test	**Instrument D (D-05, D-06):** 100% progress retained after process kill	**P1**
F-09	**ACC-06 (60% Yes):** Accessibility gap for hearing-impaired learners; missing visual cues.	**Gen Obj 1 / O6** (Pediatric Inclusion)	**NFR-ACC-01**	`ArticulationCue` Composable (Say It); sound captions kept in code, not shown (rev. 3.4)	**TC-ACC-01:** Mouth-shape cue present for 26/26 letters	**Instrument D (D-09):** Inspection of articulation cues	**P2**
F-10	**ACC-07 (4 of 5 Yes):** One caregiver reported difficulty for learners with motor challenges (tight corner toggles).	**Gen Obj 5 / O6** (Ergonomics)	**NFR-ACC-02**	`PlayItDimens` 64 dp tokens, `LessonScaffold`	**TC-ACC-02:** Automated layout audit confirming all targets ≥ 64 dp	**Instrument D (D-07):** Physical layout verification	**P2**
F-11	**Teacher Feedback:** Teacher T-2 requested advanced reading expansion track.	**Long-term Roadmap**	**Future Scope (P3)**	Data-driven Chapter Model (`Chapter 8+`)	N/A (Deferred to Post-Capstone)	N/A	**P3**
F-12	**Adviser Review:** Modeling sequence lacked formal composition spec and replay mechanism.	**HI-2** (Sequence Spec)	**FR-02**	`AudioComposer`, `LessonScript`	**TC-SCR-01:** Runtime sequence execution ≤ 15 s	**Instrument D (D-11):** Sequence timing verification	**P0**

Finding ID	Empirical Validation Evidence & Stakeholder Voice	Capstone Objective / SMART Goal	Refactored Req. ID	SDD v2.0 Design Component	Target STD Test Case(s)	Round 2 Field Verification Target	Priority
F-13	**Adviser Review:** Synthetic TTS creates schwa when prompt is a lone letter.	**HI-1** (Audio Pipeline)	**NFR-AUD-01**	Kokoro-82M / Chatterbox Pipeline, ElevenLabs clips (AM, TUB, YAM, /i/), Asset Manifest	**TC-AUD-03:** Manifest gate verification check	**Gate 3 Audit:** Expert phoneme verification	**P0**
F-14	**Adviser Review:** Learning model must guarantee child can succeed without an adult co-present.	**HI-3, SI-3** (Autonomy)	**NFR-IND-01, FR-03**	`TutorPolicy`, Spoken Carriers, 10s Idle Re-prompt	**TC-IND-01 to 03:** Spoken instruction & idle prompts	**Silent Observer Protocol:** ≥ 85% unprompted completion	**P1**
F-15	**Adviser Review:** Echoing an immediate model shows imitation, not true phonemic mastery.	**SI-4** (Spaced Recall)	**FR-NEW-REC**	`LearnerModel`, Review Scheduler	**TC-REV-01 to 03:** Warm-up & end-of-session checks	**Instrument B:** ≥ 70% recall retention rate	**P1**
F-16	**Proposal Contradiction:** Proposal's parent override conflicts with child independence.	**SI-1** (ASR Fairness)	**NFR-ASR-01**	Per-letter foil grammars (no fixed confidence threshold), prompt ladder (no hearts)	**TC-ASR-06:** False reject ≤ 15% on child voices	**Field Pilot:** Zero parental overrides needed	**P1**

4.1 Automated Test Coverage of Refactored Requirements (RTM v3.3, checked 2026-10-14)

The table records which automated tests now verify each requirement refactored during the sprint (branch `refactor/hear-say-it` at `0320385`, the code of debug APK H; `./gradlew testDebugUnitTest`: 433 tests, 0 failures, 6 font-scale checks skipped on the 360x640 profile by design). It complements, and does not replace, the field verification targets in §4.

Req. ID	Implemented components (code)	Automated test classes	Still pending (not automatable)
FR-02	`HearItSequenceBuilder`, `HearItViewModel`, `AudioResolver` (released held /m/ and /s/, and the short /i/)	`HearItSequenceBuilderTest`, `HearItViewModelTest`, `AudioResolverTest`, `HearItScreenFocusTest` (one "Sound: /m/" while playing; the play button keeps its colour)	Gate 3 teacher audit of every phoneme clip; Chapter 1's short /a/ and the letters of Chapters 2–7 still play the MVP clips
FR-03	`MicStatus`, `MicButton`, `SayItViewModel` (spoken corrections, `onScreenHidden`), `TutorPolicy`, `SayItFeedbackCopy`	`MicStatusTest`, `SayItViewModelTest`, `TutorPolicyTest`, `SayItFeedbackCopyTest`, `LayoutMatrixTest` (`sayIt_mouthCueBesideMic`: the feedback is one plain line under the mic)	Round 2 hesitation measure (Instrument B / D-10); RMS-driven ripple (planned)
FR-05	`FindItScreen`, `FindItGrid`, `GridGenerator`, `HeartManager`	`FindItViewModelTest`, `GridGeneratorTest`, `PictureAssetsTest`, `LayoutMatrixTest` (`findIt_wholeGridVisible`, `findIt_fontScale13`)	—
FR-13	`BLEND_IT_WORD_SEEDS` (`DatabaseModule`), `BlendItWordSelector`, `BlendItScreen`, `BlendItCard`; `AudioResolver` (released AM, TUB, YAM clips, audio release `2026-10-11-words`) and `VoiceCreditLine` (the Parent Zone credit for the ElevenLabs clips)	`BlendItWordSeedsTest`, `BlendItWordSelectorTest`, `BlendItViewModelTest`, `BlendItCardLayoutTest`, `LayoutMatrixTest` (`blendIt_tilesVisible`), `AudioResolverTest`, `VoiceCreditTest`, `BlendItTileDealTest` (tiles never dealt solved)	Teacher decodability audit (Instrument A Part 2); teacher audit of the AM, TUB, YAM clips; pictures for AM, TUB, YAM, ZIP
NFR-ASR-01	`SpeechValidator` (word-mode judge, foil grammars, error types), `SayItViewModel` (judges wrong partials only on a final result)	`SpeechValidatorTest`, `SayItViewModelTest`	Agreement with teachers ≥ 80% and false rejects ≤ 15% (Instruments B and C)
NFR-ACC-01	`ArticulationGroup`, `ArticulationCue` in Say It from the second miss (9 mouth-shape pictures, image release `2026-10-10-mouth`; none in Hear It since card 28c), `AudioPlayer.playSequence(onItemStart)`	`ArticulationGroupTest`, `ArticulationCueTest`, `HearItViewModelTest`, `SayItViewModelTest` (`secondMiss_showsMouthCue`), `PictureAssetsTest` (`everyMouthCueIsReleased`), `LayoutMatrixTest` (`hearIt_stepsAboveNext`, `sayIt_mouthCueBesideMic`)	Teacher check of the letter-to-mouth-shape pairs; D-09 inspection

Req. ID	Implemented components (code)	Automated test classes	Still pending (not automatable)
NFR-ACC-02	`PlayItDimens`, `LessonScaffold`, `GummyContainer` layout, `PlayItMotion`, `FeedbackEffects`, reduced-motion handling	`DimensTest`, `GummyContainerLayoutTest`, `LayoutMatrixTest` (4 sizes, font scale 1.3; `map_buttonsAre64dp`, `splash_lilyFaceClearOfStart`), `TopStatsBarLayoutTest` (greeting never cut, 64 dp card and dialog close buttons), `MapTextFloorTest` (no map text under 16 sp), `SplashScreenTest` (reduced motion), `MapLayoutTest`, `FeedbackEffectsTest`, `PlayItMotionTest`, `ZeroEmojiPolicyTest`	Physical layout verification on the A21s (D-07)

5. Requirements Sign-Off & Verification Protocol

The refactored requirements specified in this SRS v3.0 have been verified against the Capstone 2 Midterm Targets. Implementation will proceed according to the three-tier verification protocol:

1. **Automated Unit & Instrumented Verification (CI/CD):** Execution of all unit tests (`./gradlew testDebugUnitTest`) mapped to test IDs `TC-AUD-*`, `TC-MIC-*`, `TC-ASR-*`, `TC-DB-*`, and `TC-TEL-*`.
2. **Pedagogical Expert Verification (Gate 3 Audit):** Formal review of remastered audio clips and decodable word banks by certified DepEd reading specialists.
3. **Round 2 Field Validation (Weeks 7–8):** Empirical execution of Silent-Observer testing across primary target cohorts to substantiate SMART objective achievement.